Torsion

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Objective: Determination of angle of twist and performing theoretical and experimental data validation of torsion equation

Apparatus:

Apparatus is consists of work panels for holding the specimen and is provided with a weight hangers and weighing pans for loading the shaft in pure torsion. A rotating chuck with Angular measurements are used to measure the twist of the shaft.

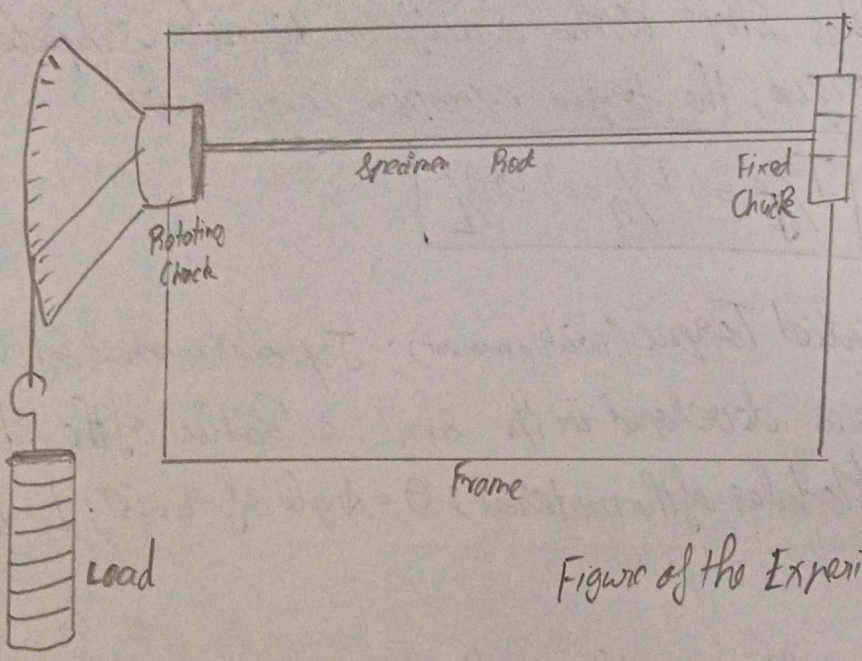


Figure of the Experimental Setup.

Theory:

A Shaft is a slender member that is primarily subjected to twisting. It is commonly used to transmit mechanical power from one location to another. When torque is applied to a shaft, shear stress develop across its cross-section. Understanding the magnitude and distribution of these stresses is important to ensure

that the shaft can withstand the applied torque without failure

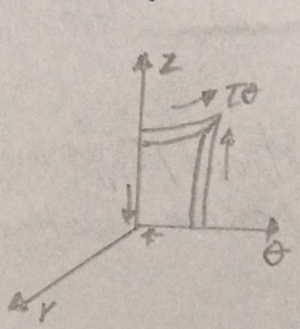


Figure showing shear stress distribution

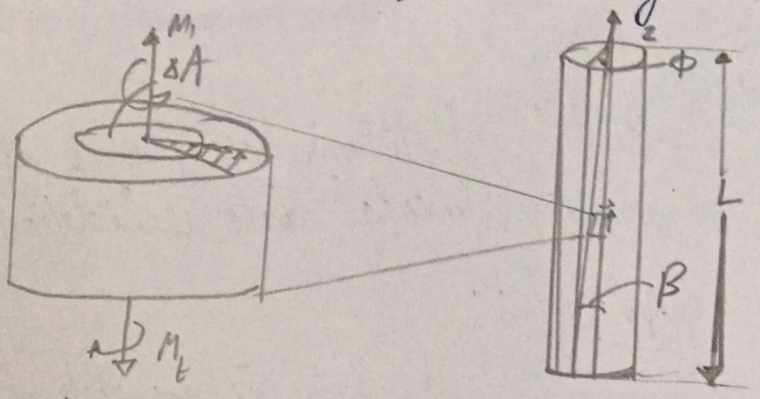


Figure showing the definition of Angle of Twist

In certain applications, shafts can also act like torsional springs, where their rotational stiffness is an important factor. In such cases, the relationship between the applied torque and the resulting angular deformation needs to be studied.

For a circular shaft with a uniform diameter subjected to a constant torque, the torque equation is given by,

$$\boxed{T/J = \tau/r = G\theta/L}$$

where,

T = Applied Torque (Twisting moment); J = polar moment of Inertia

τ = Shear stress developed in the shaft; r = Radius of the shaft

G = Shear Modulus of the material; θ = Angle of Twist; L = Length of Shaft

Procedure:

1. Measure the shaft dimensions and fix it securely in the stock
2. Attach weight hanger and set it zero.
3. Measure Shaft Length (L)
4. Add weights in 100g increments to 500g, recording the angle of twist each time.
5. Gently tap the work panel after each loading to minimize friction.
6. Calculate Angle of Twist from the reading and repeat the experiment with other materials.

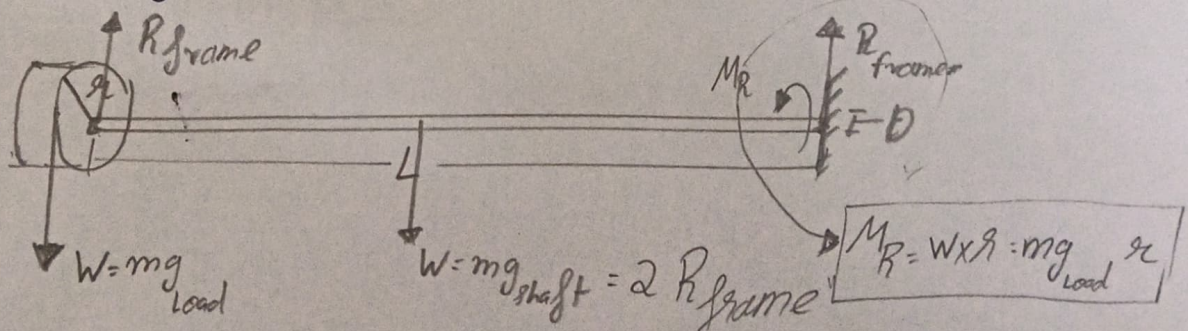
Observation:

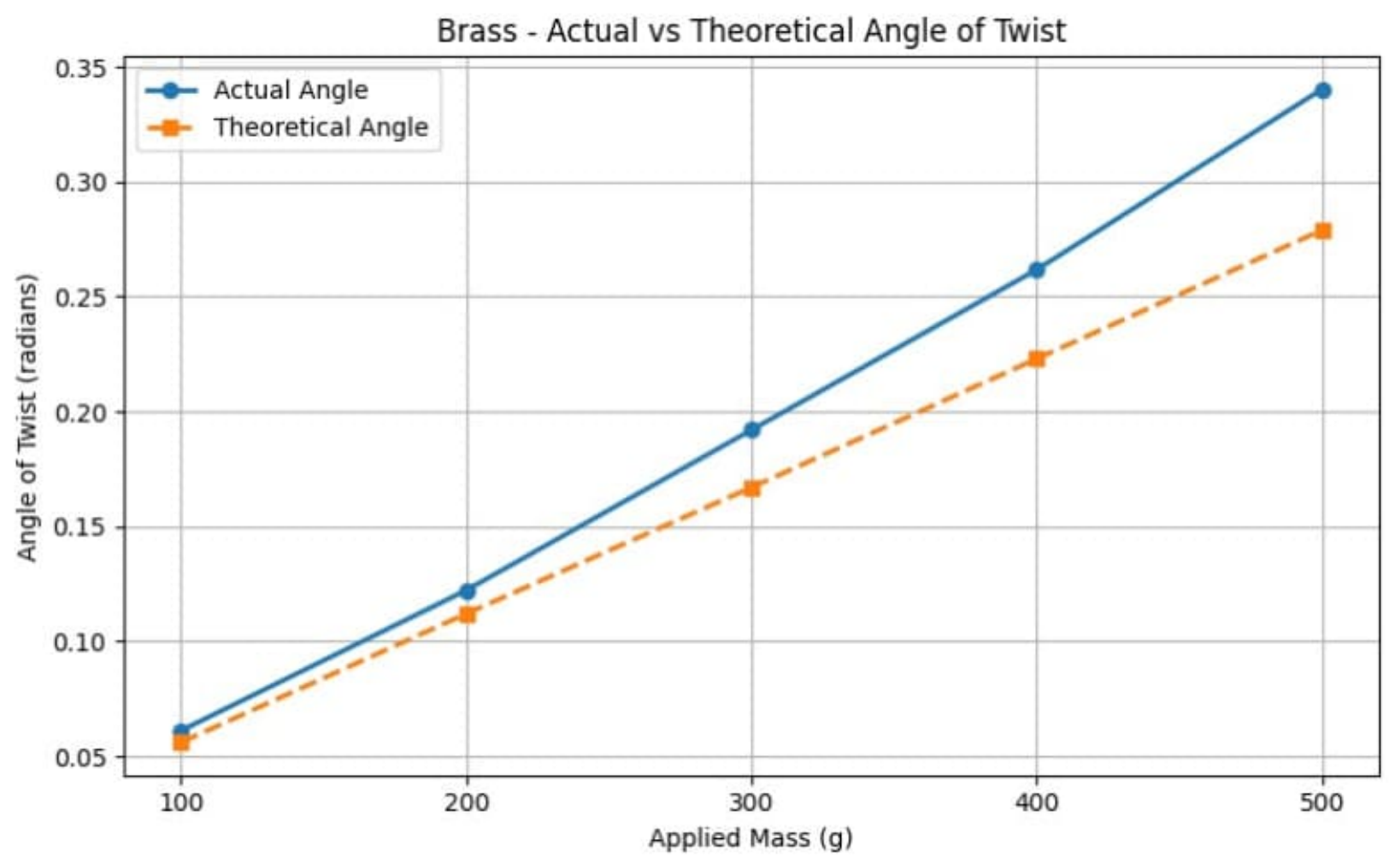
Material	Load (g)	Load (w)	Torque (Nm)	Actual Twist Angle (radians)	Theoretical Twist Angle (rad)
Brass G=38GPa	100	0.981	0.05886	0.0611	0.056
	200	1.962	0.11772	0.1222	0.112
	300	2.943	0.17658	0.1920	0.167
	400	3.924	0.23544	0.2618	0.223
	500	4.905	0.29430	0.3403	0.279
Steel G=79.6GPa	100	0.981	0.05886	0.0262	0.027
	200	1.962	0.11772	0.0524	0.053
	300	2.943	0.17658	0.0873	0.080
	400	3.924	0.23544	0.1134	0.107
	500	4.905	0.29430	0.1484	0.133
Aluminium G=25.5GPa	100	0.981	0.05886	0.0873	0.083
	200	1.962	0.11772	0.1745	0.166
	300	2.943	0.17658	0.2618	0.250
	400	3.924	0.23544	0.3665	0.333
	500	4905	0.29430	0.4712	0.416

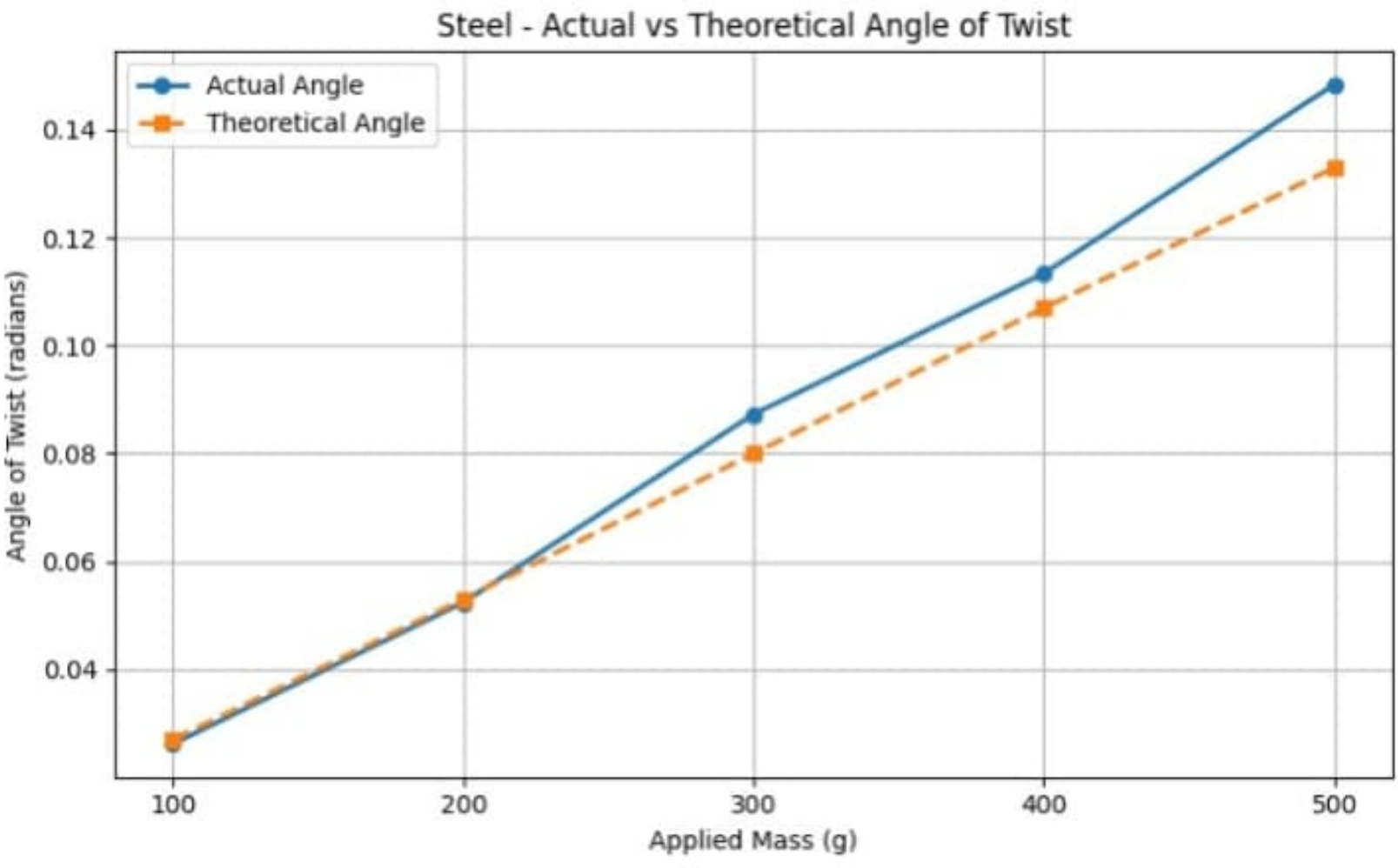
D = 3.2mm, L = 37.1cm, → J = 1.0294×10^{-11} m^4 [Shaft Dimensions]

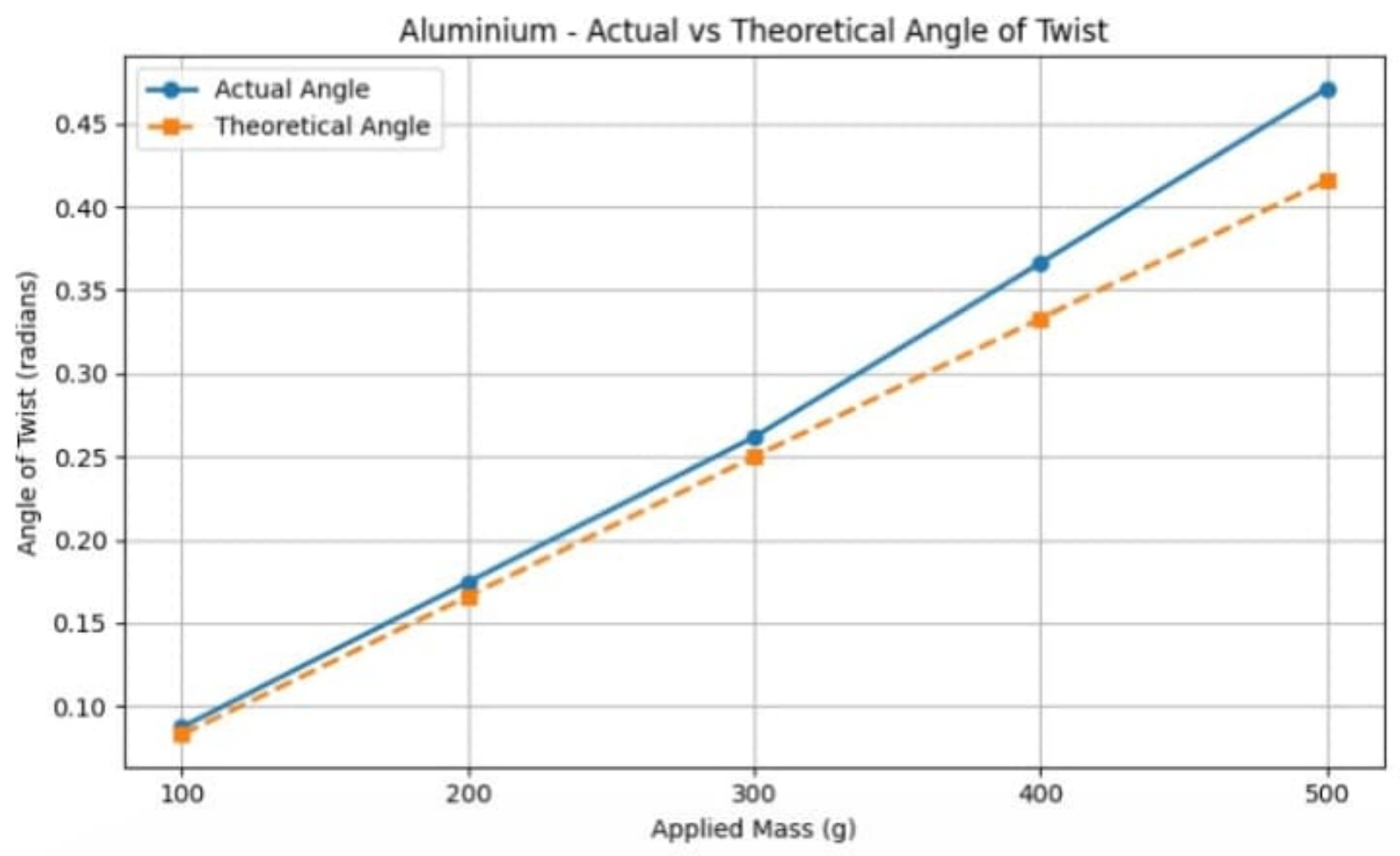
Results:

1) Free body Diagram:




Brass - Actual vs Theoretical Angle of Twist
Actual Angle
Theoretical Angle
Angle of Twist (radians)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
100
200
300
400
500
Applied Mass (g)


Steel - Actual vs Theoretical Angle of Twist
Actual Angle
Theoretical Angle
Angle of Twist (radians)
0.04
0.06
0.08
0.10
0.12
0.14
100
200
300
400
500
Applied Mass (g)


Aluminium - Actual vs Theoretical Angle of Twist
Actual Angle
Theoretical Angle
Angle of Twist (radians)
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
100
200
300
400
500
Applied Mass (g)

Discussion:

1) How is shear modulus related to young's modulus:

Relationship between G & E is $G = E/2(1+\nu)$ where ν is Poisson's Ratio.

2) Verify the relationship between torque and Angle of Twist

According to torsion Equation, $T \propto \theta$ for a given shaft. The experimental results confirm this relationship

3) How does shaft length affect the angle of twist under constant load?

The Angle of Twist is directly proportional to the shaft length. A larger shaft undergoes greater twisting under same torque. Possible errors include friction, improper loading and irregular fastening.